

Magnetic multipoles in altermagnets

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& Materials Science

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Magnetic Phases



Paramagnetic
Space Group

Irreducible representation
(irreps)

Complete list of ways symmetry
can be broken

Magnetic Space Group

- **SOC** turned on
- Focus on $q=0$ (Γ), no unit cell enlargement
- Irreps Γ_g^- time reversal odd (-), inversion even (g)
- **Collinear** configurations, 2 mag. ions per primitive UC



Not just (anti-)ferromagnetism!

Altermagnetism – Reciprocal Space

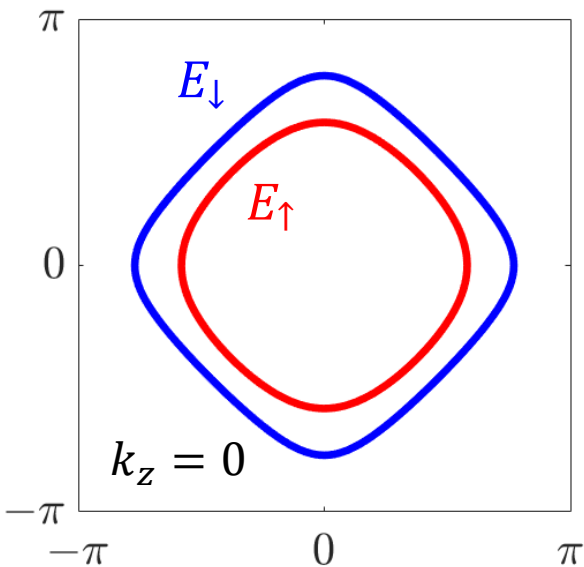
Many kinds of predicted spin splittings:

Fernandes et al., PRB **109** (2024)

Roig et al., PRB **110** (2024)

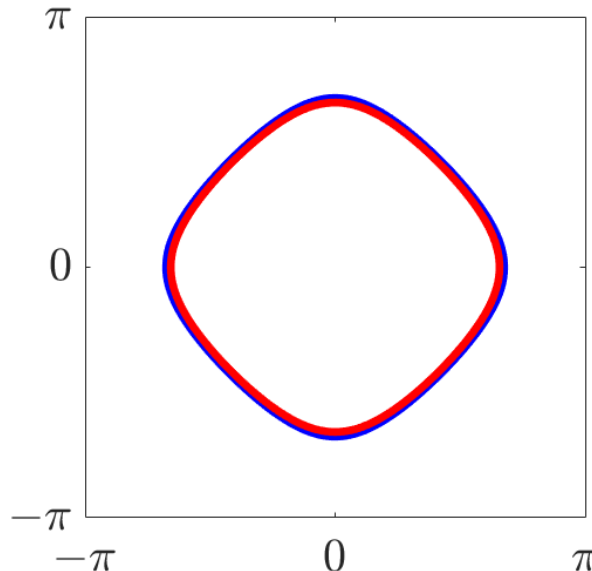
Ferromagnetism

Dipole $\sim \sigma_z$



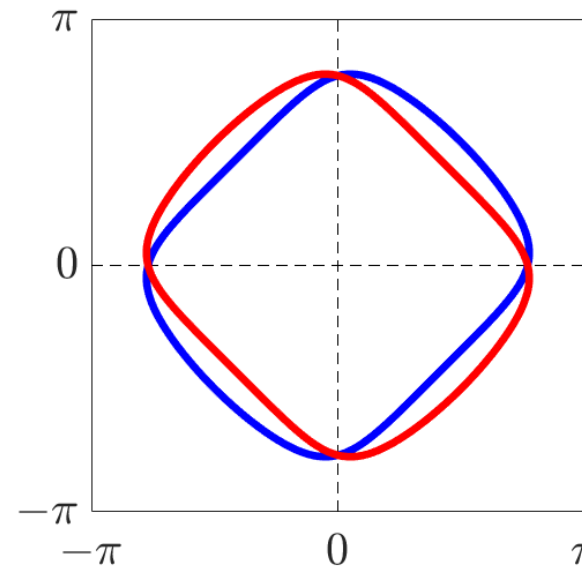
Anti-ferromagnetism

No multipole



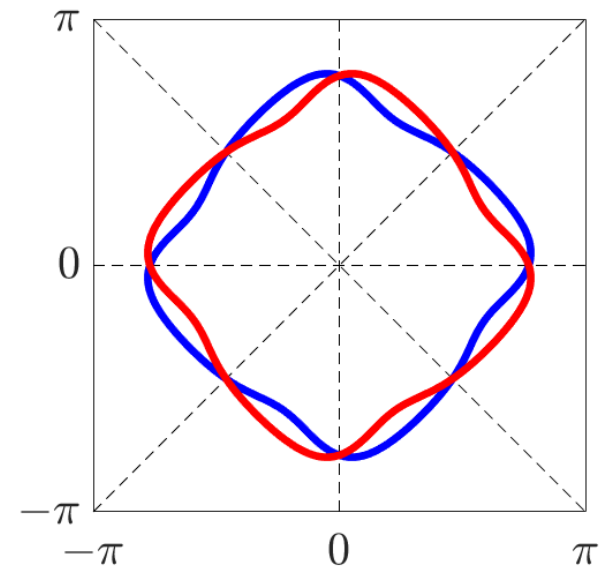
d-wave Altermagnetism

Octupole $\sim k_x k_y \sigma_z$



g-wave Altermagnetism

32-pole $\sim k_x k_y (k_x^2 - k_y^2) \sigma_z$



What about real space?

Altermagnetism – Real Space

Inside a sphere S_R of radius R centered on an atom, expand the spin density onto the tesseral harmonics:

The diagram illustrates the expansion of the spin density $\mathcal{M}_i(\mathbf{r})$ inside a sphere of radius R centered on an atom. The expansion is shown as a sum of terms:

$$\mathcal{M}_i(\mathbf{r}) = \alpha_0 s + \alpha_1 p + \alpha_2 d + \dots$$

The terms are represented by 3D visualizations:

- $\mathcal{M}_i(\mathbf{r})$: A sphere with a central black dot and six red and blue question marks distributed on its surface, representing an unknown spin density distribution.
- s : A sphere with a central black dot and a uniform red color, representing the s-harmonic (spherically symmetric).
- p : A sphere with a central black dot and two lobes (one red, one blue) oriented along a vertical axis, representing the p-harmonic.
- d : A sphere with a central black dot and four lobes (two red, two blue) in a clover-like pattern, representing the d-harmonic.

Altermagnetism – Real Space

Inside a sphere S_R of radius R centered on an atom, expand the spin density onto the tesseral harmonics:

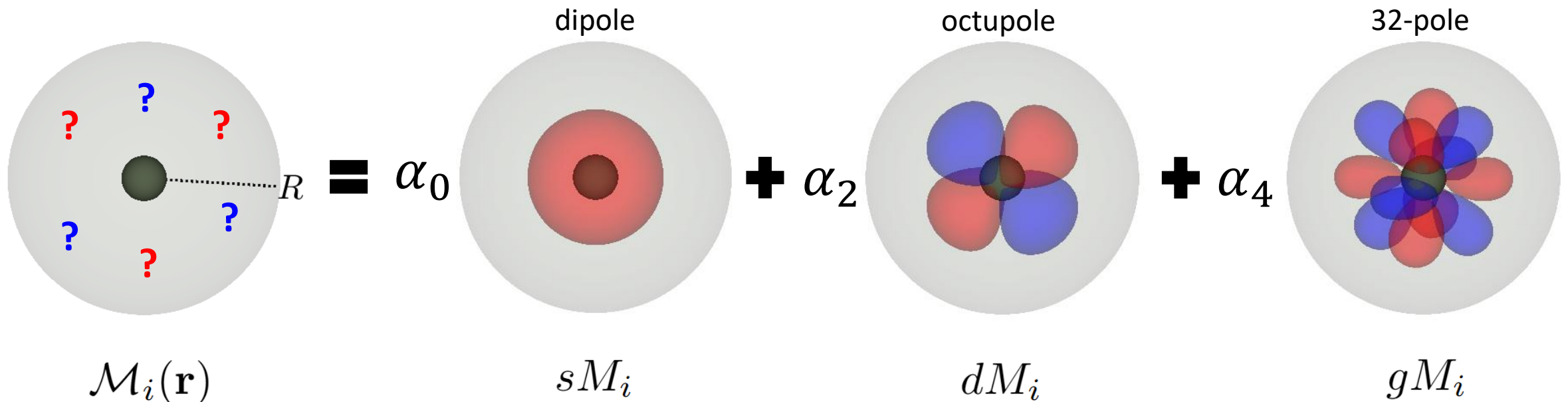
$$\mathcal{M}_i(\mathbf{r}) = \sum_{\ell,m} \alpha_{\ell,m}^i E_{\ell,m}(\mathbf{r}) M_i$$

expansion coefficients

basis vectors

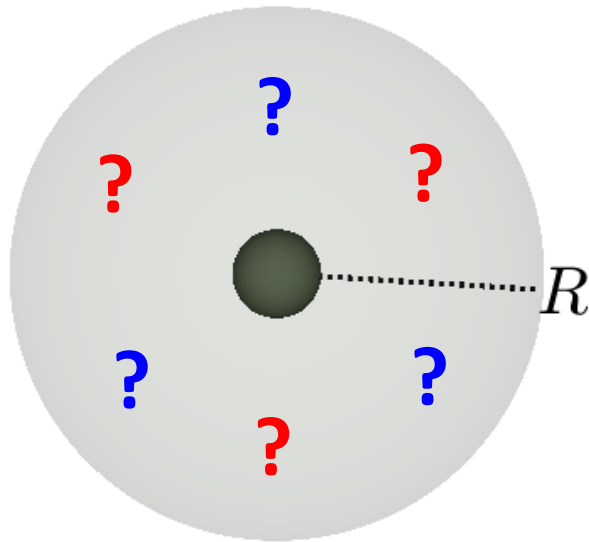
spin density

tesseral harmonics



(Atoms at inversion-even sites)

Altermagnetism – Real Space



$$\mathcal{M}_i(\mathbf{r})$$

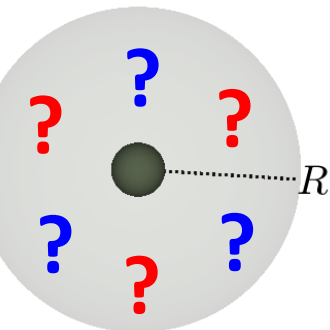
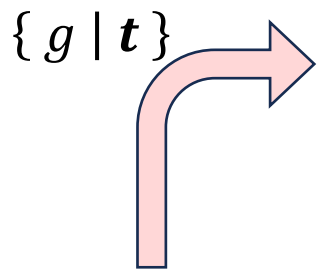
$$\mathcal{M}_i(\mathbf{r}) = \sum_{\ell,m} \alpha_{\ell,m}^i E_{\ell,m}(\mathbf{r}) M_i$$



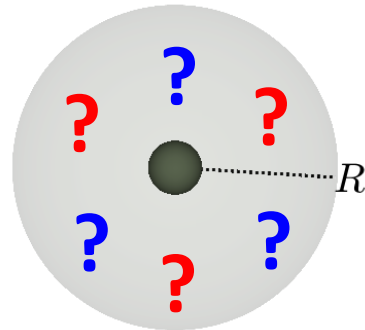
The local **site symmetry** of the atom dictates which terms are allowed in this sum for individual atoms

Altermagnetism – Real Space

Non-symmorphic
symmetry



$\mathcal{M}_i(\mathbf{r})$



$\mathcal{M}_i(\mathbf{r})$

$$\mathcal{M}_i(\mathbf{r}) = \sum_{\ell,m} \alpha_{\ell,m}^i E_{\ell,m}(\mathbf{r}) M_i$$



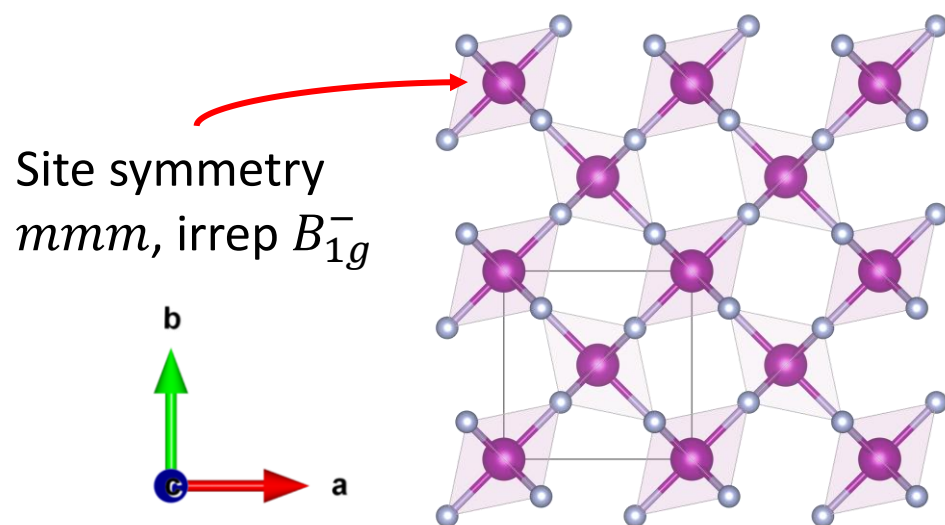
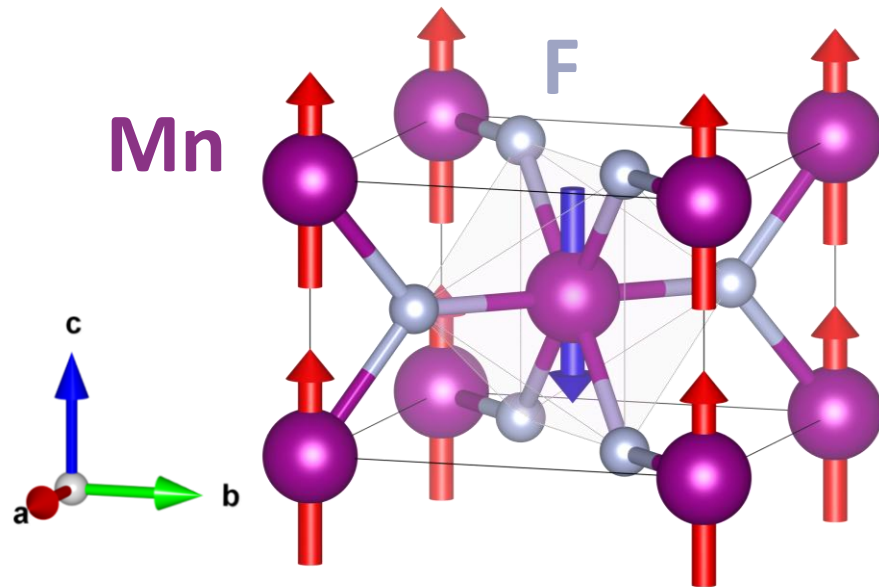
The local **site symmetry** of the atom dictates which terms are allowed in this sum for individual atoms



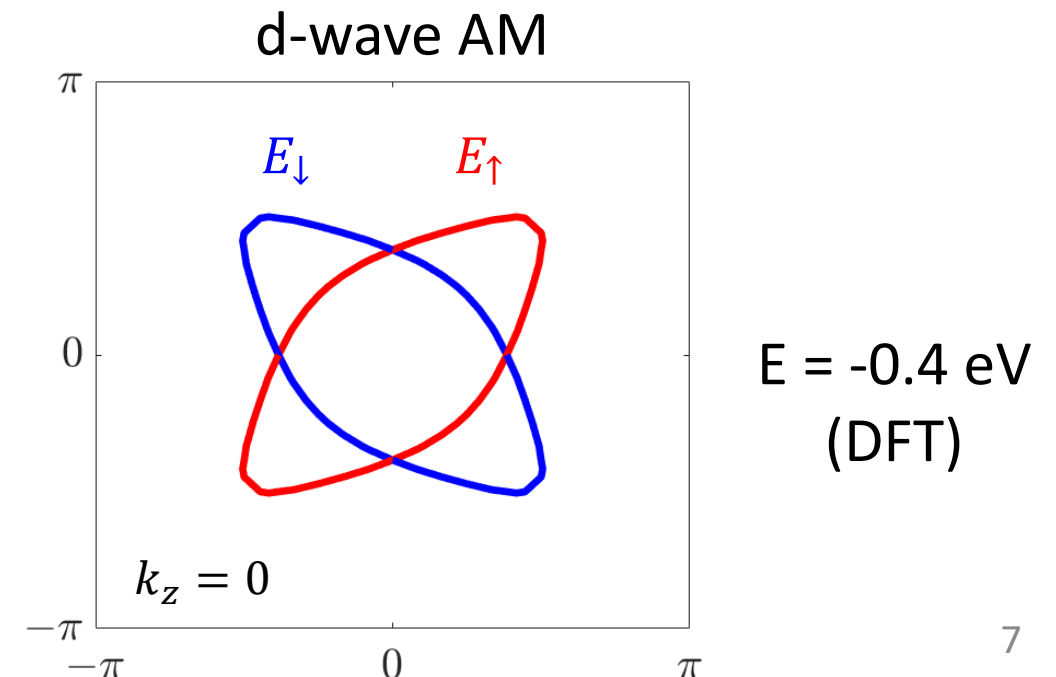
The order parameter (**paramagnetic SG irrep**) determines the relative phase between multipoles at different atoms (e.g. **ferroic** or **anti-ferroic** order)

Rutile structure – MnF_2

S. Hayami et al., JPSJ **88** (2019)
 L.D. Yuan et al., PRB **102** (2020)
 L. Šmejkal et al., Sci. Adv. **6** (2020)



$$P4_2/mnm.1' + m\Gamma_2^+ = P4'_2/mnm'$$



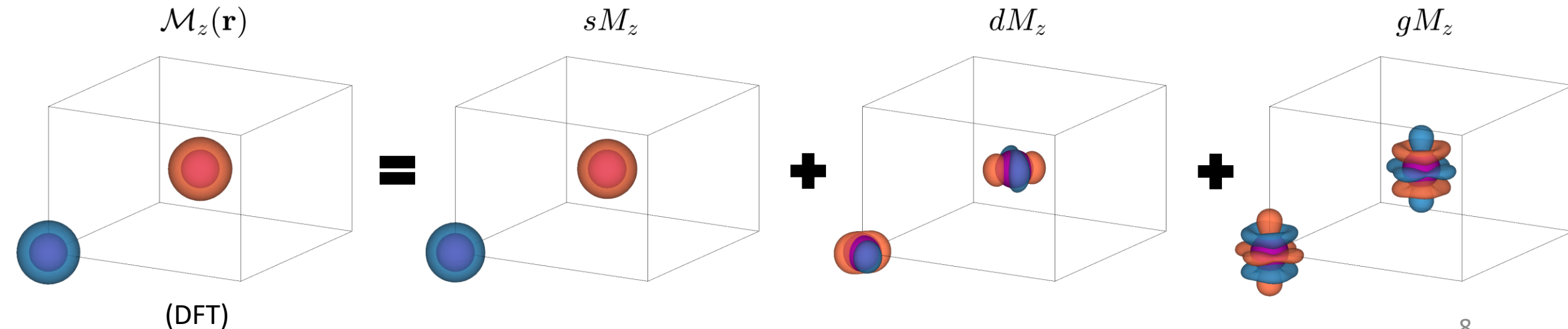
Altermagnetism in MnF_2

Enforcing the **site** symmetry B_{1g}^- of mmm :

	m_{001}	$m_{1\bar{1}0}$	m_{110}
A_g	1	1	1
B_{1g}	1	-1	-1
B_{2g}	-1	1	-1
B_{3g}	-1	-1	1



$$\{ s, d_{z^2}, d_{xy}, g_{xyz^2}, g_{x^2y^2}, g_{z^4} \} M_z$$



Altermagnetism in MnF_2

Enforcing the **SG irrep**
 $m\Gamma_2^+$:

Enforcing the **site** symmetry B_{1g}^- of mmm :

Anti-ferroic (σ_z)
 even under 4_{001}

$$\{ s, d_{z^2}, g_{x^2y^2}, g_{z^4} \} M_z \sigma_z$$

$$\{ d_{xy}, g_{xyz^2} \} M_z \sigma_0$$

Ferroic (σ_0)
 odd under 4_{001}



	$\{4_{001} \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\}$	$\{m_{100} \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\}$	$\{m_{110} 0, 0, 0\}$
$m\Gamma_1^+$	1	1	1
$m\Gamma_2^+$	-1	1	-1
$m\Gamma_3^+$	1	-1	-1
$m\Gamma_4^+$	-1	-1	1

Sub-lattice space (pseudo-spin $\frac{1}{2}$)

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Altermagnetism in MnF_2

Enforcing the **SG irrep**
 $m\Gamma_2^+$:

Enforcing the **site** symmetry B_{1g}^- of mmm :

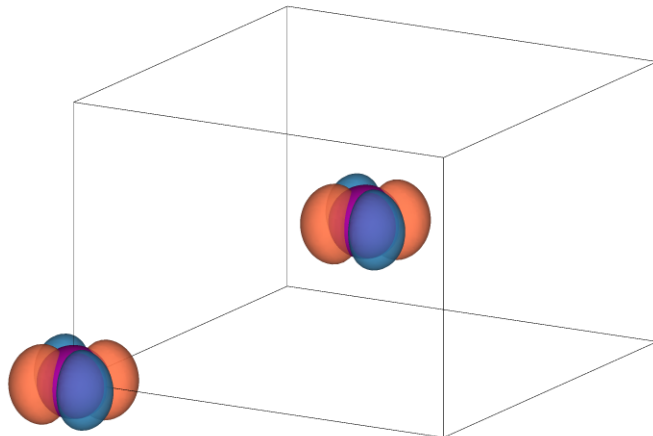
Anti-ferroic (σ_z)
 even under 4_{001}

Ferroic (σ_0)
 odd under 4_{001}

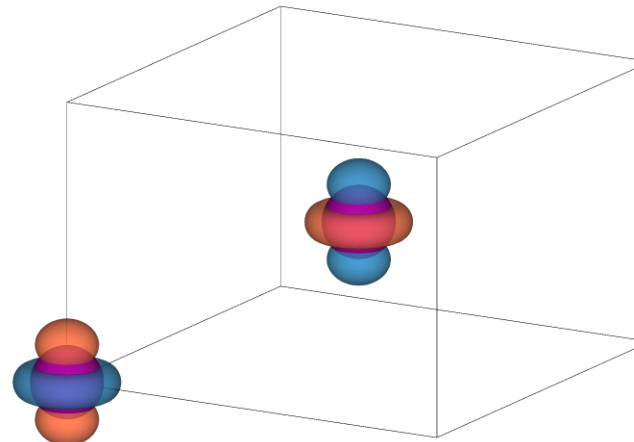
$$\{ s, d_{z^2}, g_{x^2y^2}, g_{z^4} \} M_z \sigma_z$$

$$\{ d_{xy}, g_{xyz^2} \} M_z \sigma_0$$

$$d_{xy} M_z$$



$$d_{z^2} M_z$$



Bhowal and Spaldin,
 PRX **14**, (2024)

Altermagnetism in MnF_2

Enforcing the **SG irrep**
 $m\Gamma_2^+$:

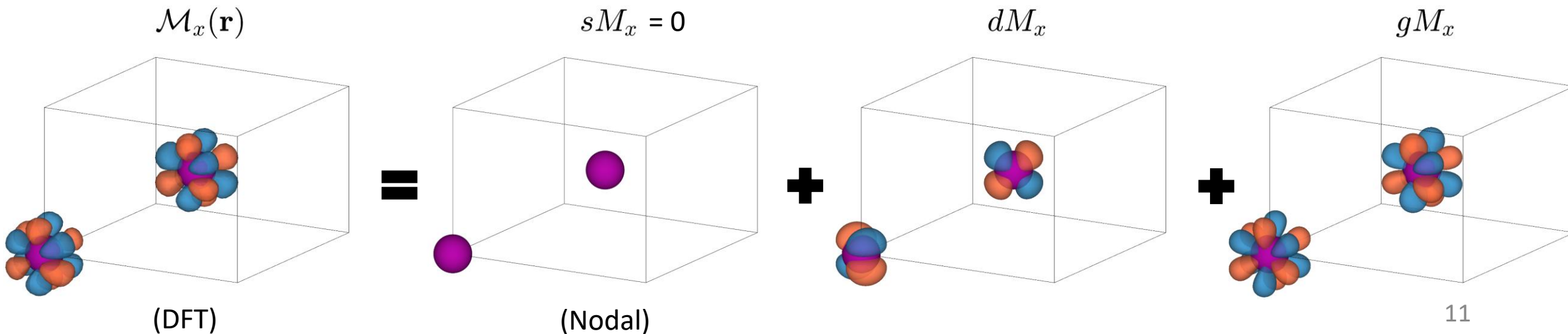
Enforcing the **site** symmetry B_{1g}^- of mmm :

$$\{ d_{yz}, g_{yz}^3, g_{yz}(y^2 - 3x^2) \} M_x \sigma_0 + (x \rightarrow y)$$

$$\{ d_{xz}, g_{xz}^3, g_{xz}(x^2 - 3y^2) \} M_x \sigma_z + (x \rightarrow y)$$

Ferroic (σ_0)
 odd under 4_{001}

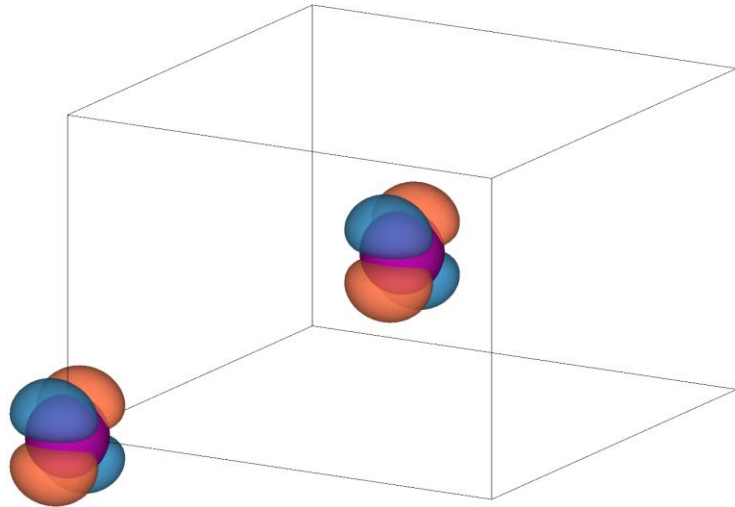
Anti-ferroic (σ_z)
 even under 4_{001}



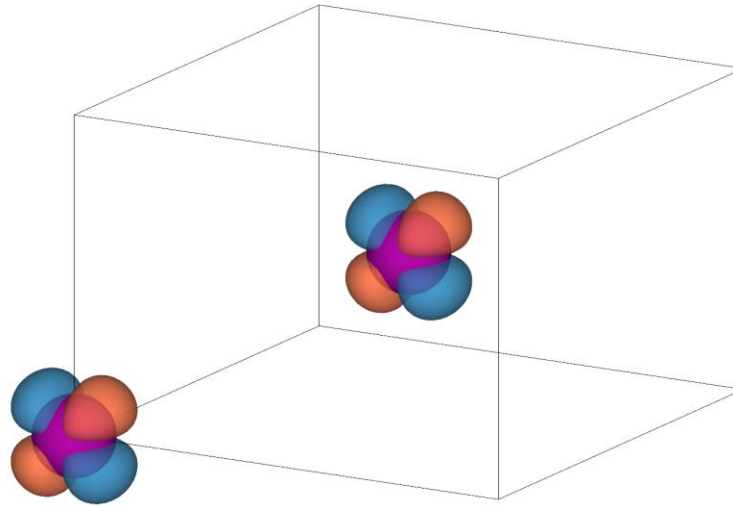
Altermagnetism in MnF_2

Lowest-order **ferroic** multipoles:

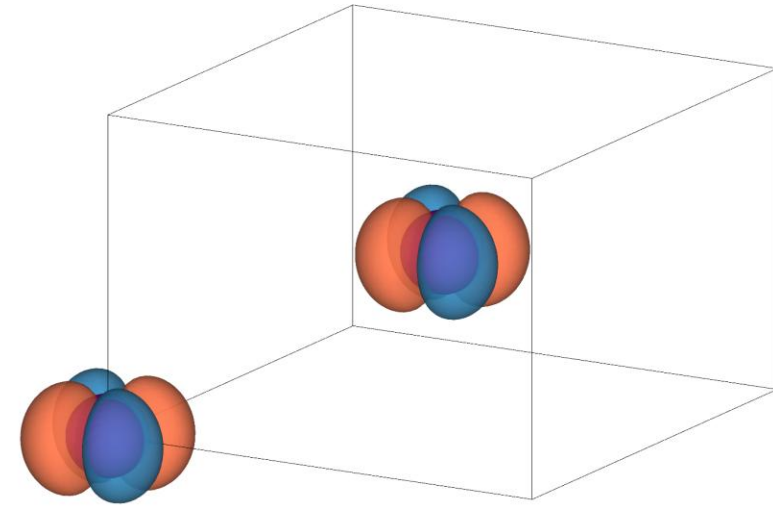
$$d_{yz}M_x$$



$$d_{xz}M_y$$



$$d_{xy}M_z$$

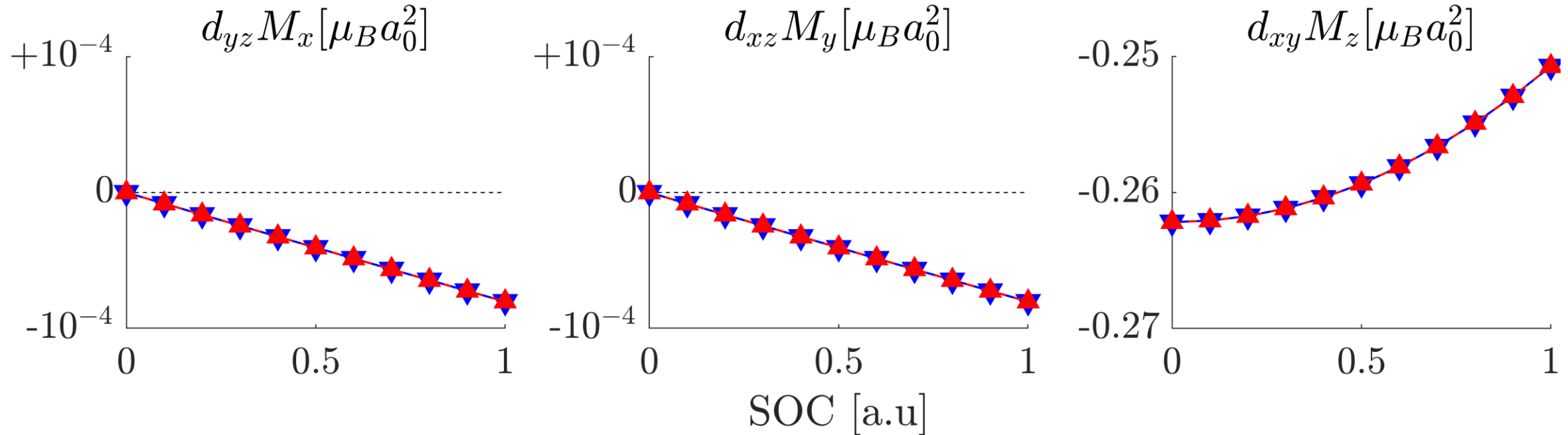


Tensors $a_e[V^2]V$ allowed matrix elements (e.g. Piezomagnetism):

$$\Lambda_{ij} = \begin{bmatrix} 0 & 0 & 0 & \lambda_{yzx} & 0 & 0 \\ 0 & 0 & 0 & 0 & \lambda_{xzy} & 0 \\ 0 & 0 & 0 & 0 & 0 & \lambda_{xyz} \end{bmatrix}$$

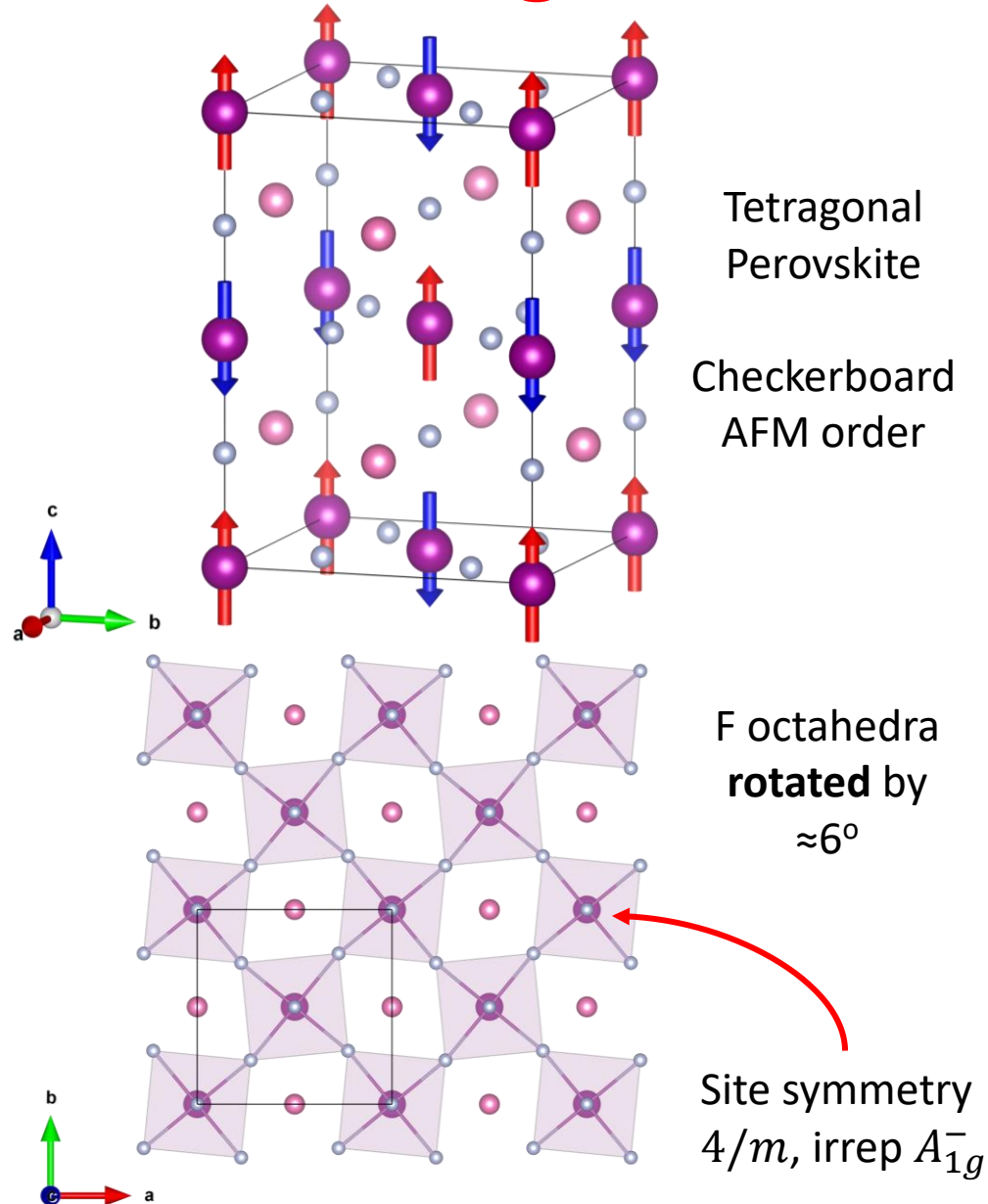
Altermagnetism in MnF_2

DFT results as a function of SOC scaling: $\mathcal{H}_{ij}^{\text{SOC}} = \lambda_{\text{SOC}} \alpha^2 \int \phi_i^*(\vec{r}) \left(\frac{1}{r} \frac{\partial V}{\partial r} \vec{L} \cdot \vec{S} \right) \phi_i^*(\vec{r}) d^3r$

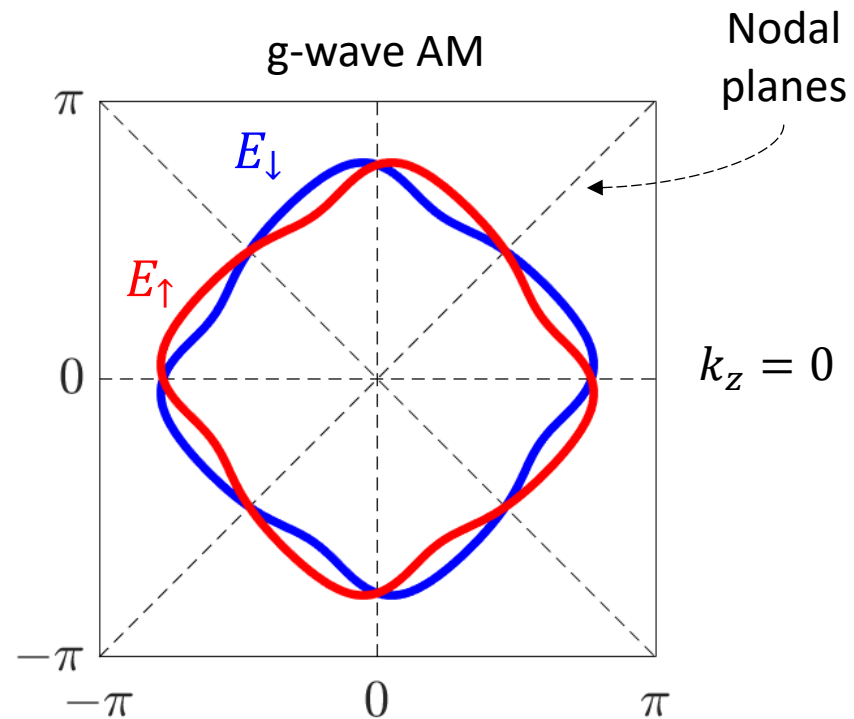


Tensors $\text{ae}[V^2]V$ allowed matrix elements (e.g. Piezomagnetism): $\Lambda_{ij} = \begin{bmatrix} 0 & 0 & 0 & \lambda_{yzx} & 0 & 0 \\ 0 & 0 & 0 & 0 & \lambda_{xzy} & 0 \\ 0 & 0 & 0 & 0 & 0 & \lambda_{xyz} \end{bmatrix}$

Altermagnetism in KMnF_3



$$I4/mcm.1' + m\Gamma_1^+ = I4/mcm.1$$

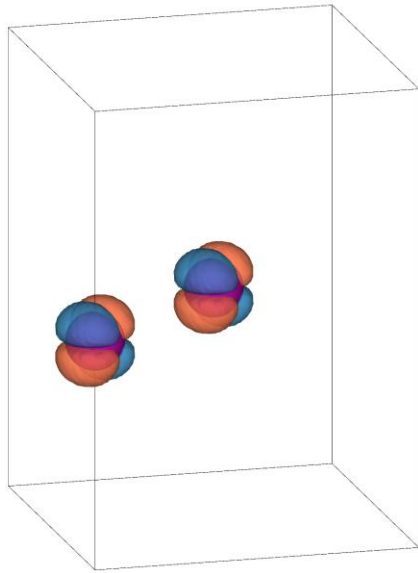


M. Naka et al.,
npj Spintronics **3** (2025)

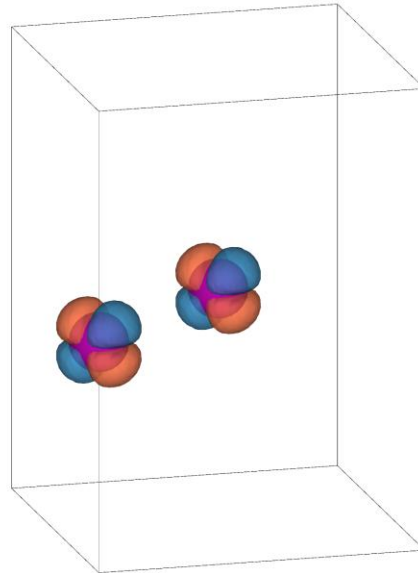
Altermagnetism in KMnF_3

Lowest-order **ferroic** multipoles:

$$d_{yz}M_x$$



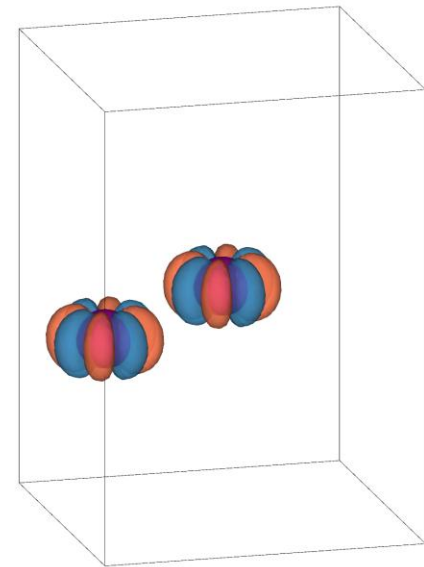
$$d_{xz}M_y$$



g-wave AM

$$g_{xy}(x^2-y^2)M_z$$

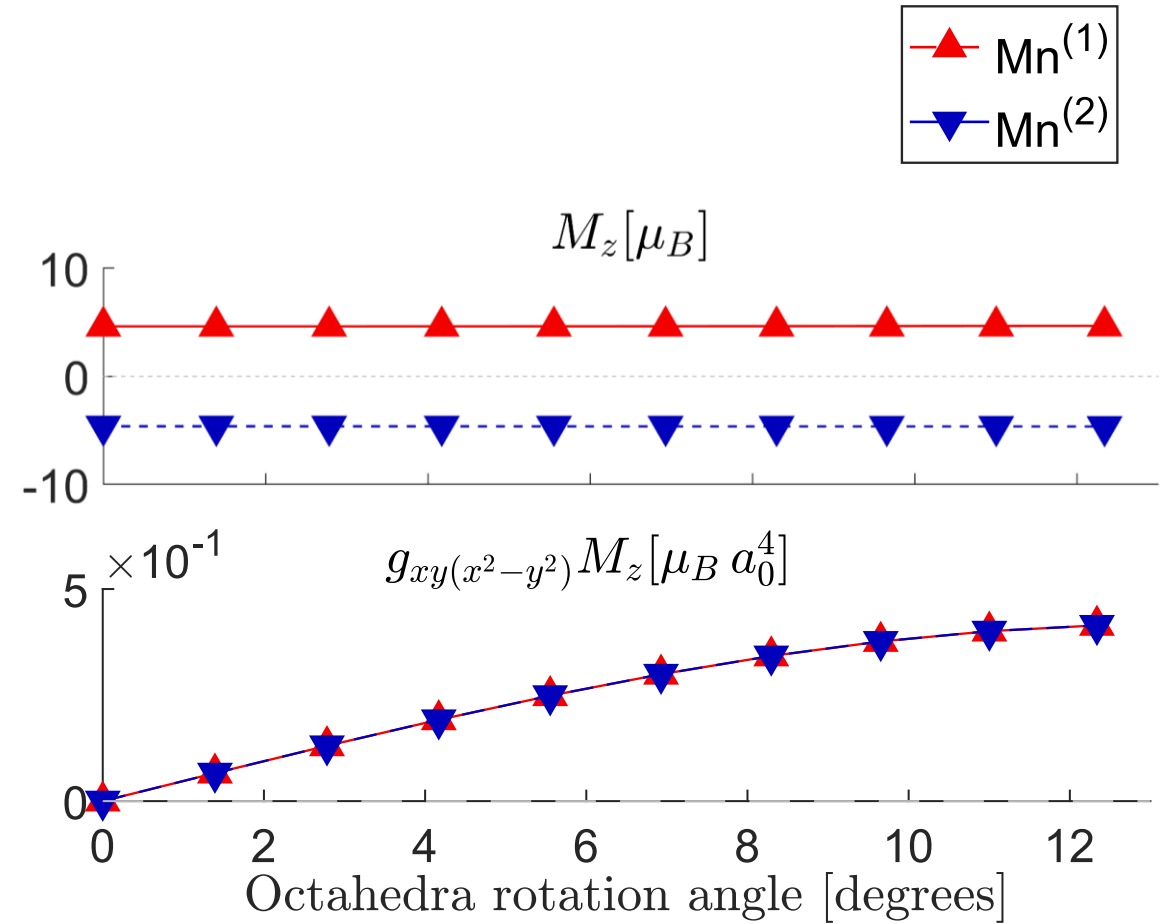
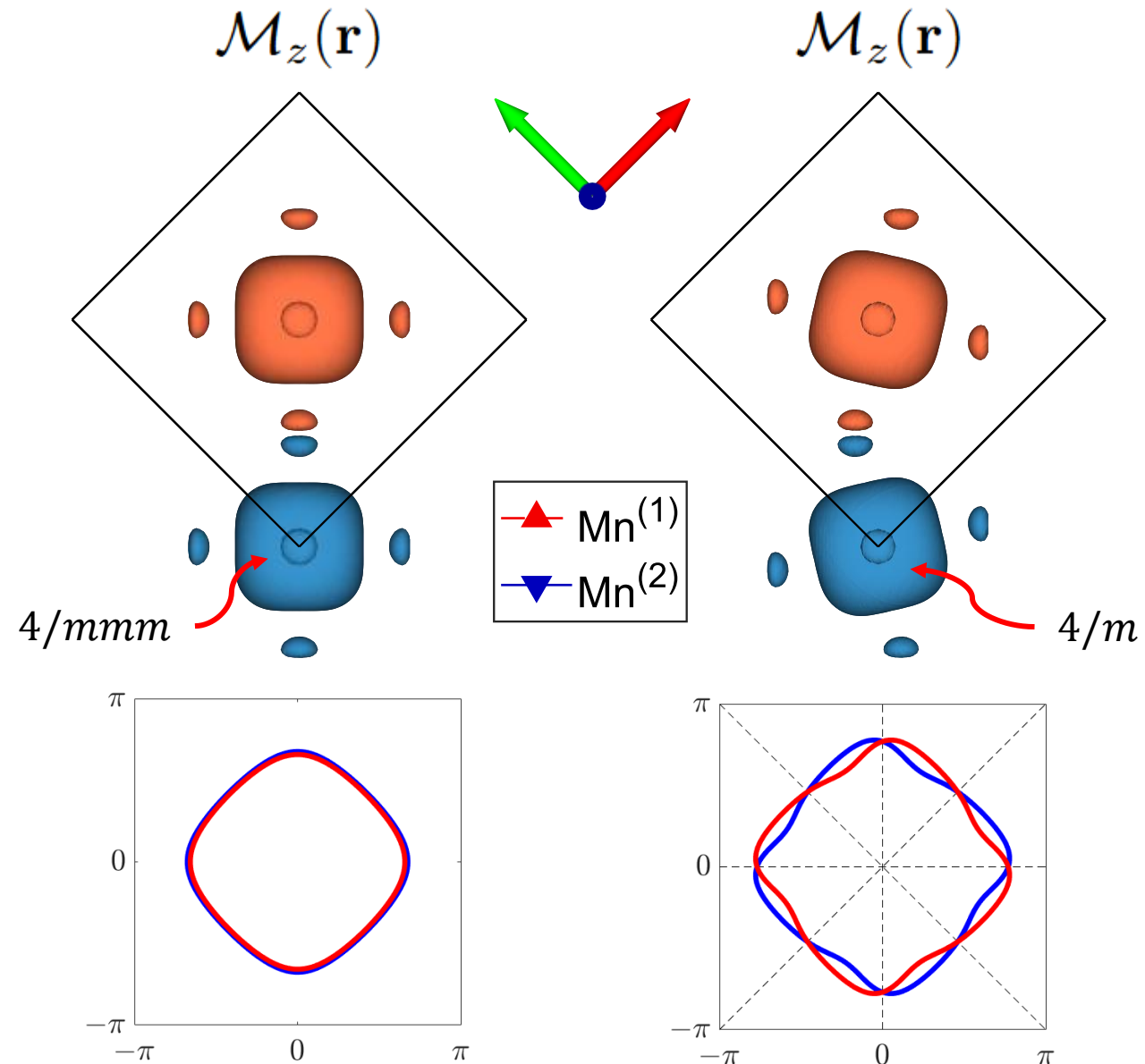
Tensors
ae[[V²][V²]]V



Tensors ae[V²]V allowed matrix elements (e.g. Piezomagnetism):

$$\Lambda_{ij} = \begin{bmatrix} 0 & 0 & 0 & \lambda_{yzx} & 0 & 0 \\ 0 & 0 & 0 & 0 & \lambda_{xzy} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

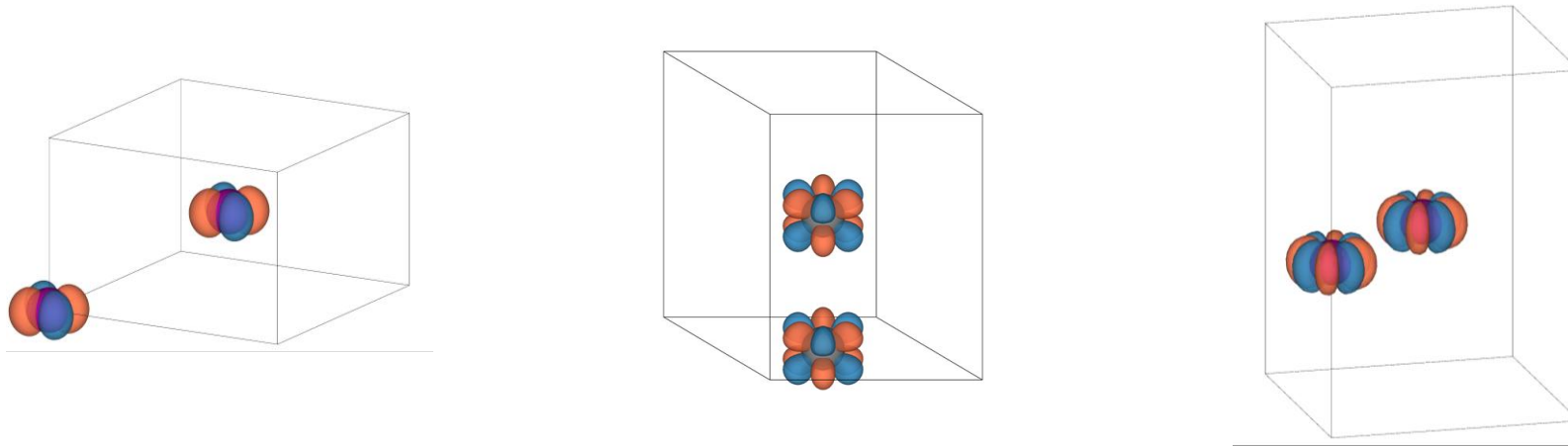
Tuning AM with structural DOFs



Experimental angle $\approx 6^\circ$

Conclusions

- Classified relative phase of magnetic multipoles through the paramagnetic SG irreps
- Showed their dependence on SOC



Similar analysis could be applied to:

- Non-collinear phases
- Systems with more atoms per unit cell
- Inversion-odd magnetic multipoles